

The Inner Compass: Small Language Models for Context-Aware Enterprise Assistance

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Abstract—Enterprise IT environments involve a wide range of systems, configurations, and technical documents, which can make it difficult for employees to find the right support when an issue occurs. Conventional L0–L1–L2–L3 support processes generally depend on manual escalation and static knowledge bases, which may increase the time needed to identify and resolve problems. This paper presents an enterprise support and monitoring system based on a Small Language Model (SLM). The system uses organizational information from sources such as network diagrams, server configurations, database schemas, operational manuals, and logs to understand employee queries and provide relevant support.

The approach combines Retrieval-Augmented Generation (RAG), natural language understanding, and criticality detection to distinguish between critical and non-critical issues. Relevant procedures and previous resolutions are processed and indexed so that information related to a particular query can be retrieved when required. The SLM then uses this retrieved context to generate step-by-step guidance for the reported problem. Permission controls and human-in-the-loop mechanisms are included for situations where a recommended action could affect a user's device or the enterprise environment. User interactions can also provide information for improving the knowledge base and identifying recurring support issues.

The system follows a local-first deployment approach, allowing organizational information to remain within the enterprise network while reducing dependence on external cloud services. In addition to providing direct assistance, the system records issue patterns, resolution outcomes, and knowledge gaps that can help identify areas where documentation or employee training may need improvement. By combining SLM-based assistance with enterprise knowledge retrieval, criticality detection, and controlled actions, the proposed approach aims to provide context-aware IT support while maintaining security, interpretability, and administrative control.

Index Terms—Enterprise Support Systems, Small Language Models (SLM), Retrieval-Augmented Generation (RAG), IT Operations Monitoring, Knowledge Management, Natural Language Processing, Criticality Detection, Human-in-the-Loop Automation

I. INTRODUCTION

Modern enterprise IT environments comprise heterogeneous components, including networks, servers, databases, virtualization layers, and containerized workloads. These systems are supported by extensive documentation such as runbooks, standard operating procedures, knowledge-base articles, architecture documents, and incident reports. However, much operational knowledge remains tacit and depends on the expe-

rience of support teams. When employees encounter technical problems, organizations commonly use an L0–L1–L2–L3 support model, where issues are escalated according to severity. This process can increase coordination overhead and the time required for resolution.

The growing adoption of cloud-native architectures, microservices, and dynamic workloads has further increased this challenge. Infrastructure is more distributed, configurations change frequently, and dependencies between services are harder to track. Support teams must therefore interpret logs and metrics from multiple sources while maintaining consistent operational knowledge. Quickly locating relevant procedures and converting expert knowledge into reusable guidance remain important challenges.

Conventional approaches such as manual ticketing, static knowledge bases, and rule-based workflows are effective for repetitive tasks but depend on users knowing where to find and interpret the required information. When knowledge is distributed across multiple repositories, issues may be escalated unnecessarily or take longer to resolve. AIOps and ITSM tools improve alerting and incident management, but may provide limited contextual assistance. Similarly, generic LLM assistants may lack reliable access to organization-specific procedures and may recommend actions without sufficient consideration of security and permissions.

To address these limitations, this paper investigates the use of Small Language Models (SLMs) as enterprise support assistants. The proposed system uses a local-first SLM to interpret employee queries, identify critical and non-critical issues, and retrieve relevant information through Retrieval-Augmented Generation (RAG). For routine issues, the system provides step-by-step guidance based on documented procedures and previous resolutions, while potentially critical issues can be directed toward appropriate escalation paths.

The system also supports enterprise monitoring by recording issue patterns, resolution outcomes, and knowledge gaps. Permission controls and human-in-the-loop mechanisms are incorporated to ensure that sensitive operations, such as software installation or configuration changes, require explicit user or administrator approval. By operating within the enterprise environment, the system also provides greater control over organizational information and reduces dependence on external cloud services.

The main contributions of this work are as follows:

- 1) An SLM-based enterprise support architecture for first-line technical assistance.
- 2) Integration of natural-language understanding, criticality detection, and RAG over enterprise documentation.
- 3) A framework for using support interactions to identify recurring issues and knowledge gaps.
- 4) A monitoring mechanism for analyzing support patterns and potential training requirements.
- 5) A permission and human-in-the-loop model for maintaining security and administrative control.

By combining SLM-based assistance, enterprise-specific retrieval, criticality detection, and controlled execution, the proposed system provides a practical framework for context-aware IT support while maintaining security, privacy, and administrative oversight.

II. LITERATURE SURVEY

Small Language Models (SLMs) have gained attention as efficient alternatives to large language models for applications requiring local processing, lower computational cost, reduced latency, and improved data privacy. These characteristics are particularly relevant to enterprise environments containing sensitive configurations, internal procedures, and incident information. However, their limited reasoning capability can create challenges for complex, multi-step tasks.

A. Small Language Models for Efficient Deployment

Minaee et al. survey SLM architectures, training methods, compression techniques, and applications, highlighting knowledge distillation, pruning, quantization, and parameter-efficient fine-tuning as approaches for reducing model size while maintaining performance [1]. Wang et al. similarly report that smaller models are effective for specialized and well-defined tasks, while noting limitations in generalization and reasoning [2]. Zhang et al. show that SLM performance depends not only on model size but also on architecture, training quality, and task alignment [3]. Lu et al. demonstrate the potential of SLMs for resource-constrained deployment, supporting their use in local enterprise environments [4]. Pham et al. further demonstrate the use of compact models for on-device document understanding and question answering through SlimLM [5].

B. SLMs for Agentic and Decision-Making Tasks

SLMs face additional challenges when used for agentic tasks requiring multi-step reasoning and decision-making. Fukuda, Nozue, and Oishi investigate SLM agents for continuously updating ICT systems and propose task decomposition, dynamic prompt construction, and retrieval of reusable operational trajectories [6]. Their experiments on ALFWorld and WideEnet demonstrate that suitable retrieval and exemplar selection can improve SLM performance on multi-step operations. However, the approach depends on sufficient operational examples and may introduce additional computational overhead. These limitations highlight the need for

carefully maintained knowledge sources in enterprise support applications.

C. SLMs for Enterprise Knowledge Assistance

Enterprise support requires models to understand organization-specific terminology and retrieve relevant procedures from distributed knowledge sources. An SLM combined with an enterprise knowledge base can provide a practical interface for accessing runbooks, manuals, ticket histories, and configuration documents. Rather than storing all organizational knowledge in model parameters, the system can retrieve relevant information at inference time and provide focused context to the model. This approach is particularly useful for privacy-sensitive environments, where internal technical information should remain within the organization's infrastructure.

D. Monitoring and Continuous Improvement

Enterprise support systems can also use interaction data to identify recurring issues, missing documentation, outdated procedures, and employee training requirements. However, user interactions should not be directly incorporated into the knowledge base without validation. Support-team verification, document versioning, access control, and administrator approval are required to maintain reliable organizational knowledge. Regularly updating verified knowledge can therefore improve support relevance while reducing the risk of propagating incorrect information.

E. Research Gap

Existing studies establish the benefits of SLMs for efficient and privacy-preserving deployment, while agent-based research demonstrates their potential for multi-step ICT operations. However, limited work combines SLM-based query understanding, enterprise knowledge retrieval, criticality detection, employee-facing troubleshooting, monitoring, and permission control within a single framework.

This paper addresses this gap through an SLM-based enterprise support system that retrieves organization-specific knowledge, distinguishes routine issues from potentially critical incidents, provides context-aware guidance, and maintains human oversight for sensitive operations. The system also records support interactions to identify recurring issues and knowledge gaps while keeping enterprise information within a controlled environment.

III. METHODOLOGY

The proposed enterprise IT support system combines Qwen3-8B with Retrieval-Augmented Generation (RAG) to provide context-aware assistance for employee IT issues. The methodology includes dataset preparation, model adaptation, enterprise knowledge retrieval, response generation, criticality assessment, permission control, and audit logging.

A. Dataset Preparation

The Qwen3-8B model is adapted using an IT Operations (IT-Ops) dataset containing user queries and technical responses across monitoring, networking, hardware, security, Windows/server administration, cloud platforms, Microsoft 365, Linux, and databases.

The data is preprocessed by removing malformed records, unsuitable responses, identity-specific references, duplicate question-answer pairs, and classification-only examples. The resulting dataset contains 75,654 examples, divided into 60,523 training, 7,565 validation, and 7,566 test samples.

B. Conversational Data Formatting

The processed question-answer pairs are formatted using the Qwen tokenizer and chat template. Each example represents an employee query followed by its corresponding support response:

$$\text{User Query} \rightarrow \text{Assistant Response} \quad (1)$$

The formatted conversations are tokenized and provided to the fine-tuning process.

C. Qwen3-8B Adaptation Using QLoRA

Qwen3-8B is adapted using Quantized Low-Rank Adaptation (QLoRA) with 4-bit quantization. The pretrained model parameters remain frozen while trainable LoRA adapters are introduced into selected layers. The adapted weight matrix is represented as

$$W' = W + \Delta W \quad (2)$$

where the parameter update is expressed as

$$\Delta W = BA \quad (3)$$

where A and B are low-rank trainable matrices. In the implemented system, 43,646,976 of 8,234,382,336 parameters are trainable, corresponding to approximately 0.5301%.

D. Supervised Instruction Fine-Tuning

The adapted model is trained using supervised instruction tuning, where each IT-support query is paired with its target response. The causal language-modeling objective is

$$\mathcal{L} = - \sum_{t=1}^T \log P(y_t | y_{<t}, x) \quad (4)$$

where x is the input query, y_t is the target token, and T is the response length. Only the LoRA parameters are updated during training.

E. Enterprise Knowledge Retrieval

The fine-tuned model is integrated with a RAG pipeline to access enterprise-specific information. Enterprise documents are embedded using Sentence Transformers with the all-MiniLM-L6-v2 model and indexed using FAISS.

For an employee query, the same embedding model converts the query into a vector, which is compared with the indexed document vectors. The most relevant document segments are retrieved along with metadata such as domain, environment, document type, and source file.

F. Context-Aware Response Generation

The retrieved information is combined with the original query and provided to Qwen3-8B as contextual input. The retrieval process is represented as

$$C = R(q, K) \quad (5)$$

where q is the employee query, K is the enterprise knowledge base, and C is the retrieved context.

The response is then generated as

$$y = M(q, C) \quad (6)$$

where M represents the adapted Qwen3-8B model and y is the generated response. This allows updated enterprise information to be used without retraining the language model whenever documentation changes.

G. Criticality Assessment

The system assesses the reported issue to distinguish routine requests from potentially critical incidents. Non-critical issues can receive troubleshooting guidance, while potentially critical issues can be directed toward escalation or additional review. This decision layer helps prevent high-impact issues from being treated as routine support requests.

H. Permission and Human-in-the-Loop Control

The model provides recommendations but does not independently execute restricted operations. Actions such as software installation, configuration changes, or service modifications require explicit user or administrator authorization. High-impact operations are subject to human review before execution, maintaining administrative control over enterprise resources.

I. Deployment and User Interaction

Employees interact with the system through a natural-language interface. The query is processed through model inference, knowledge retrieval, criticality assessment, and permission control before the response or escalation path is presented.

The system is organized into modular components for user interaction, Qwen3-8B inference, knowledge retrieval, decision control, and auditing, allowing individual components to be updated independently.

J. Audit Logging and Monitoring

The system records relevant support information, including queries, generated responses, recommendations, and approved actions. These records can be analyzed to identify recurring IT issues, knowledge gaps, and outdated documentation, supporting improvements to enterprise support processes.

K. Overall Framework

The complete methodology integrates:

- 1) **Qwen3-8B SLM** for natural-language understanding and response generation.
- 2) **QLoRA and 4-bit quantization** for parameter-efficient model adaptation.
- 3) **RAG** using `all-MiniLM-L6-v2` embeddings and FAISS.
- 4) **Context-aware generation** using retrieved enterprise information.
- 5) **Criticality assessment** for identifying issues requiring escalation.
- 6) **Permission and human-in-the-loop controls** for sensitive operations.
- 7) **Audit logging** for traceability and support monitoring.

Overall, the framework combines domain-adapted Qwen3-8B with enterprise knowledge retrieval while separating automated assistance from sensitive operational actions. This enables context-aware IT support while maintaining security, authorization, and administrative oversight.

IV. SYSTEM OVERVIEW

The proposed enterprise support system is designed to generate context-aware IT assistance based on employee-reported issues and organizational knowledge. The system combines domain-adapted Small Language Model (SLM) capabilities with Retrieval-Augmented Generation (RAG) to provide accurate, efficient, and scalable support guidance. The architecture emphasizes **personalization, efficiency, interpretability, and operational safety**, making it suitable for deployment within enterprise IT environments.

The system follows a modular design, allowing integration of additional features such as incident tracking, knowledge-base updates, user feedback mechanisms, and advanced monitoring dashboards.

A. System Modules

The system is structured around four primary modules.

1) User Interaction Module: The module collects employee queries in natural language describing IT issues or support requests. It performs input validation and basic sanitization while supporting real-time interaction through a web or chat interface. When available, user context such as role, department, and affected systems is also captured.

2) SLM Adaptation Module: The module loads the pre-trained base language model using 4-bit quantization and applies QLoRA-based parameter-efficient fine-tuning on IT-support conversations. It generates a domain-specific adapter

for IT operations guidance while keeping more than 99% of the base model parameters frozen.

3) RAG Knowledge Module: This module maintains an indexed repository of enterprise documents, including runbooks, SOPs, knowledge-base articles, architecture diagrams, and past incident reports. Documents are converted into embeddings and stored in a vector database for semantic retrieval. Relevant knowledge chunks are retrieved based on the employee query and available context. Access control is applied to restrict information according to user roles, while document metadata and timestamps can be used to identify outdated or superseded procedures.

4) Support Guidance Module: The module combines the employee query with retrieved RAG knowledge as context for the adapted SLM. It generates step-by-step resolution instructions based on organizational procedures, classifies issues as critical or non-critical, recommends escalation paths for potentially critical incidents, and provides source references to improve traceability.

B. System Workflow

The end-to-end workflow of the system consists of the following stages.

1) Employee Query Submission: The employee submits a natural-language description of the IT issue through the interface. The query is validated and basic information such as timestamp, user role, and affected system is captured.

2) Query Interpretation: The adapted SLM processes the query to identify the affected system, symptoms, and error descriptions. Relevant keywords and contextual information are extracted for subsequent knowledge retrieval.

3) RAG-Based Knowledge Retrieval: The query is embedded using the same embedding model used for the knowledge base. The RAG module searches the vector index for top- k semantically similar knowledge chunks. Retrieved information is filtered based on access control, relevance, and recency, while outdated or conflicting procedures can be deprioritized using document metadata.

4) Criticality Classification: The system classifies the reported issue as critical or non-critical using rule-based indicators and learned patterns. Critical issues are flagged for priority handling and potential escalation.

5) RAG-Augmented Response Generation: The adapted SLM receives the employee query together with the retrieved RAG context. It generates a response grounded in organizational knowledge, including step-by-step guidance, relevant source documents, and escalation recommendations when required.

6) Permission and Safety Checks: Actions that modify a user's device or environment require explicit permission. High-risk operations require human approval before execution, and relevant actions and recommendations are recorded for auditing.

7) Output Delivery and Feedback: The generated guidance is presented to the employee through the interface. An optional feedback mechanism can capture usefulness ratings

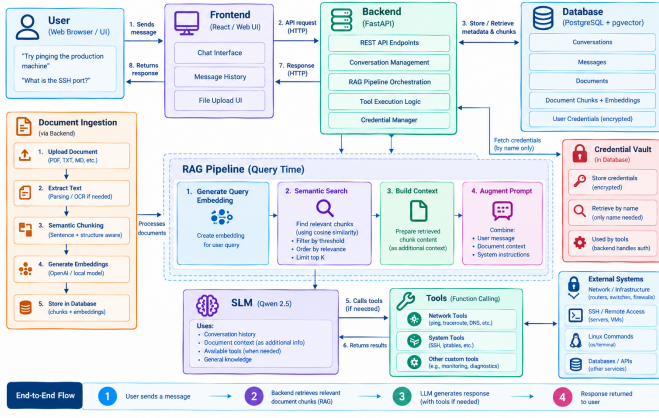


Fig. 1. Enterprise Support System Architecture with SLM and RAG

or resolution status, while interaction data can be retained for monitoring and future improvement.

C. Technical Considerations

Efficiency: QLoRA-based adaptation reduces the trainable parameters to approximately 0.53% of the base model. Four-bit quantization reduces memory requirements, while RAG allows relevant enterprise information to be retrieved on demand instead of storing all organizational knowledge within the model.

Scalability: The modular architecture supports additional knowledge sources and document types. The vector index can be updated as documents are added or revised, and separate LoRA adapters can be developed for different domains using the same base model.

Personalization: Responses are grounded in organization-specific procedures through RAG. Access control can restrict information according to employee roles, while criticality detection supports appropriate prioritization and routing.

Security and Safety: Local deployment helps keep sensitive operational information within the enterprise environment. RAG access control restricts retrieval to authorized documents, while permission enforcement prevents changes without explicit consent. Human-in-the-loop approval is used for high-risk operations, and audit logging supports traceability and accountability.

Continuous Improvement: Interaction logs can be analyzed to identify recurring issues and knowledge gaps. Feedback from employees and support staff can inform knowledge-base updates, while revised operational procedures can be incorporated into the RAG index with appropriate versioning.

D. Architecture Diagram

Figure 1 shows the overall architecture of the proposed enterprise support system.

V. CONCLUSION

This paper presented an SLM-based approach for providing IT support using organization-specific knowledge. The main

idea is to combine a domain-adapted language model with RAG so that responses are based on relevant enterprise documents rather than only on the model's stored knowledge.

The methodology uses QLoRA for parameter-efficient adaptation and includes preprocessing steps such as cleaning, deduplication, and conversational formatting. The support workflow also considers issue criticality and uses permission controls when an action could affect a user's device or the enterprise environment.

An important part of the proposed design is the use of human approval for high-risk operations. This prevents the language model from directly making sensitive changes and keeps the final decision with an authorized person. The system can also use interaction and resolution data to identify recurring problems and gaps in existing documentation.

Overall, the proposed framework provides a practical direction for using smaller language models in enterprise IT support. Its local deployment model, retrieval-based knowledge access, and controlled execution make it suitable for environments where privacy and operational safety are important.

The framework also provides a foundation for improving enterprise support processes through better knowledge management and analysis of recurring incidents. Future work can focus on evaluating the system with real-world enterprise workloads and improving its accuracy, scalability, and response efficiency.

VI. FUTURE ADVANCEMENTS

The proposed enterprise support system can be extended and enhanced in several directions to improve performance, adaptability, and user experience:

- 1) **Multi-Domain Adapter Development:** Extending the system to develop **domain-specific LoRA adapters** for different organizational units (e.g., networking, databases, security) can improve specialization and response accuracy for distinct IT domains.
- 2) **Advanced Retrieval Mechanisms:** Incorporating **hybrid retrieval strategies, re-ranking models, or graph-based knowledge representations** can capture complex relationships between documents, systems, and procedures, improving retrieval relevance.
- 3) **Personalized Feedback Learning:** Integrating **user feedback mechanisms** (resolution success, helpfulness ratings, escalation outcomes) enables the system to adapt dynamically and refine recommendations over time.
- 4) **Multimodal Knowledge Integration:** Leveraging **diagrams, network topologies, configuration snapshots, and video tutorials** can provide richer and more intuitive guidance, enhancing user understanding and engagement.
- 5) **Real-Time Operational Integration:** Connecting with **monitoring tools, ticketing systems, and configuration management platforms** allows real-time updates of support guidance based on system state, recent changes, or active incidents.

- 6) **Explainable AI Integration:** Providing **explanations for recommendations** (e.g., “based on runbook X, section Y”) increases transparency, user trust, and decision-making clarity.
- 7) **Scalability and Deployment Optimization:** Developing **lightweight and containerized implementations** enables deployment across multiple enterprise sites, improving accessibility, fault tolerance, and performance.
- 8) **Continuous Knowledge Updates:** Implementing **automated ingestion pipelines** for new documentation, updated procedures, and lessons learned from incidents ensures the RAG knowledge base remains current and accurate.
- 9) **Enhanced Security and Audit Capabilities:** Strengthening **access control, encryption, and audit logging** mechanisms supports compliance with enterprise security policies and regulatory requirements.

By incorporating these advancements, the system can evolve into a **comprehensive, adaptive, and intelligent enterprise support platform**, capable of supporting personalized IT assistance, proactive incident management, and long-term operational excellence.

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